In the Eppinger framework, **Target Specifications** translate the subjective "voice of the customer" (the needs) into objective, measurable "engineering metrics."
This step is where the project moves from the marketing domain into the engineering domain.
Each spec must have a **metric** (what we measure) and a **value** (the goal).

Translating Needs to Target Specifications

Here is how the primary needs from our previous table would look when translated into a Target Spec sheet.

Need Nos.	Metric	Importance	Units	Marginal Value	Ideal Value
2.1	Peak acoustic noise level	5	dB	< 75	< 60
1.1, 1.4	Mass of debris removed (standard test)	5	grams	> 15	> 25
3.4	Time to clean/reset device	4	seconds	< 180	< 60
3.1	Number of hands required for operation	5	count	1	1
4.3	Total mass of device (empty)	2	kg	< 3.0	< 1.5
2.2	Water/Air temperature range	4	°C	30–40	37–38
4.2	Battery life (continuous run time)	3	minutes	> 15	> 30

Key Concepts

Eppinger principles:

1. The Metric Must Be Verifiable

A spec is useless if you can't test it. Instead of "feels light" (Need 4.3), we use **"Total mass in kg."** This allows the engineering team to weigh the prototype and know immediately if they have passed or failed.

2. Marginal vs. Ideal Values

- **Marginal Value:** The "bare minimum" to be competitive in the market. If you don't hit this, the product is likely a failure.
- **Ideal Value:** The "stretch goal" that would make the product a market leader.

3. Subjective to Objective Mapping

Some needs are hard to quantify. For **Need 2.1 (Not startling the dog)**, engineers don't just measure volume (dB); they might also look at **frequency (Hz)**, because dogs are more sensitive to high-pitched whines (like those from a vacuum motor).

The "Conflict" Stage

The Trade-off - ***"If we want to hit the Ideal Value for Battery Life (30 mins), what happens to the Ideal Value for Mass (1.5 kg)?"*** Usually, more battery = more weight.

The Eppinger process forces the team to resolve these technical contradictions using a **Competitive Benchmarking** chart.

In the Eppinger (Ulrich & Eppinger) product development process, the stage immediately following the setting of Target Specifications is **Concept Generation**. While Target Specs tell the team *what* the product must achieve (the "metrics"), Concept Generation is where the team explores *how* to achieve it. This is the most creative part of the process, but Eppinger emphasizes a structured approach to ensure you don't just "brainstorm" and hope for the best.

The 5-Step Concept Generation Process

Eppinger breaks this stage down into a repeatable method that is perfect for teaching CS and Engineering students:

1. Deconstruct the Problem (Functional Decomposition)

Break the complex problem (cleaning a dog) into smaller sub-problems.

- **Example:** How do we *provide water*? How do we *agitate dirt*? How do we *dry the fur*? How do we *manage waste*?

2. Search Externally

Look for existing solutions to these sub-problems.

- **Patents:** Search for existing nozzle or vacuum technologies.
- **Lead Users:** Talk to professional groomers or vets.
- **Analogies:** How do car washes clean tires? How do surgical scrubs work?

3. Search Internally (Brainstorming)

The team generates as many ideas as possible for each sub-problem. The rule here is **"Suspend Judgment."** No idea is too expensive or weird at this stage.

4. Systematic Exploration (The Concept Combination Table)

This is the "Management" part of the process. You create a **Morphological Matrix** where you list sub-functions and various ways to solve them. You then "mix and match" solutions to create complete product concepts.

Sub-Function	Option A	Option B	Option C
Water Delivery	Integrated Tank	Sink Attachment	Ultrasonic Mist
Dirt Removal	Rotating Brushes	High-Pressure Air	Silicone "Nubs"
Drying Method	Heated Air	Microfiber Wrap	Centrifugal Spin

5. Reflect on the Solutions

The team steps back to see if there are any gaps or if certain concepts are physically impossible based on the **Target Specs** (e.g., a "Centrifugal Spin" might be too scary for the dog, violating a high-importance need).

For Engineering and CS students, this stage is a "reality check." It moves them from a single "cool idea" to a **range of architectural choices**.

For example, in a dog cleaning device:

- **Concept 1:** A "Dry" solution using UV light and specialized vacuums (Tech-heavy).
- **Concept 2:** A "Portable Power" solution using a backpack water tank and manual scrubbing (Mechanical-heavy).
- **Concept 3:** A "Smart" solution that connects to an app to track skin health while washing (Software-heavy).

The "Concept Selection" Preview

After they generate these concepts, the *next* logical step would be **Concept Selection** (using a **Pugh Matrix**), where they compare these ideas against their Target Specs to see which one "wins."

To create a **Concept Combination Table** (also known as a **Morphological Matrix**), we take the high-level function "Clean a Dog" and decompose it into its essential sub-functions.
By looking at this table, you can "trace a path" through the options to create entirely different product architectures.

Morphological Matrix: The "PawPure" Dog Cleaning System

Sub-Function	Option 1 (The "High-Tech" Path)	Option 2 (The "Mechanical" Path)	Option 3 (The "Waterless" Path)
1. Agitate Dirt	Ultrasonic Vibrations	Rotating Silicone Brushes	High-Pressure Air Stream
2. Apply Cleanser	Automated Mist Spray	Manual Squeeze Trigger	Pre-treated Wipes/Pads
3. Water Source	Internal 1L Reservoir	Direct Sink Hookup	No Water (Dry Foam)
4. Debris Removal	Integrated Vacuum Suction	Gravity/Drainage Tray	Microfiber Absorption
5. Drying Method	High-Velocity Warm Air	Centrifugal "Spin" Cup	Passive Air-Dry
6. User Interface	LED Touchscreen/App	Physical Toggle Switch	Pressure-Activated Only

You can "build" a product using only one path:

- **Group A (The "Apple" of Dog Washers):** Combine **Option 1** for all functions. This leads to a premium, quiet, automated device (Needs 2.1 and 1.2).
- **Group B (The "Budget/Reliable" Version):** Combine **Option 2** for all functions. This leads to a rugged, battery-free tool that attaches to a faucet (Need 4.1).
- **Group C (The "Traveler's" Solution):** Combine **Option 3**. This creates a portable, light-weight device for cars or hiking (Need 4.3).

The Next Step: Concept Selection (Pugh Matrix)

Once you have 3–4 "Winning Concepts" from the table above, the Eppinger process moves to **Concept Selection**. This is where they use a **Pugh Matrix** to compare new ideas against a "Datum" (usually the current market leader or a "Standard Bathtub").

The Pugh Matrix helps answer: *"Is our 'Ultrasonic' idea actually better at removing mud than a 'Rotating Brush'?"*

In the Eppinger methodology, **Concept Selection** is where the team uses a **Pugh Matrix** to narrow down the brainstormed ideas. Instead of picking a favorite based on gut feeling, you compare each concept against a **"Datum"** (the current standard) using the Customer Needs we identified earlier.
For a dog cleaning device, the most logical "Datum" to compare against is a **Standard Bathtub/Shower**.

Pugh Concept Selection Matrix: Dog Cleaning Device

Customer Need (Criteria)	Importance	**Datum: Bathtub**	Concept A: Ultrasonic Mist	Concept B: Rotating Brush	Concept C: Vacuum/Dry
1.1 Removes mud/debris	5	0	+	+	–
2.1 Low noise/Startle	5	0	+	–	–
2.4 pH Balanced/Skin Safe	5	0	0	0	+
3.1 One-hand operation	5	0	+	+	+
3.2 Minimize splash-back	4	0	+	+	++
3.4 Self-cleaning time	4	0	+	–	+
4.3 Portable/Lightweight	2	0	+	+	+
Sum Score (+)			**6**	**4**	**4**
Sum Score (0)			**1**	**1**	**1**
Sum Score (-)			**0**	**2**	**2**
Net Score		**0**	**+6**	**+2**	**+2**
Rank			**1**	**2**	**2**
Continue?			**Yes**	**Combine**	**No**

Key

- **(+):** Better than the datum.
- **(-):** Worse than the datum.
- **(0):** Same as the datum.

Key Points

1. **Objective Decision Making:** Notice that "Concept A" wins because it addresses the high-importance needs (Noise and One-hand operation) better than the others. It helps engineers avoid "falling in love" with a cool but loud mechanical brush.
2. **Combine & Improve:** Look at Concept B. It's great at removing mud (+) but fails on noise

(-). A manager might ask: *"Can we take the rotating brush from B and add the sound-dampening tech from A?"* This is what Eppinger calls **Concept Combination**.

3. **The "Losing" Concept:** Concept C (Vacuum/Dry) failed on mud removal. For a "dog cleaning device," if it can't clean mud (Importance 5), it doesn't matter how portable it is. This allows the team to "fail fast" and stop wasting resources on a flawed architecture.

The Final Step: Concept Scoring

If Concept A and B are close, you would perform a second round called **Concept Scoring**. In that round, you don't just use plus/minus; you multiply the "Importance" weight (1–5) by a performance score (1–5) to get a weighted total.